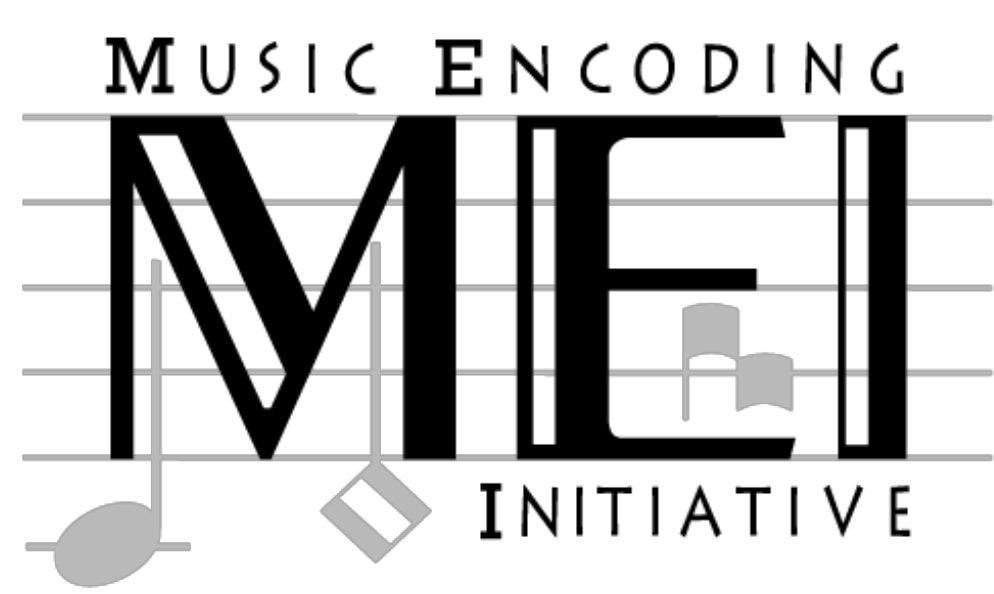


MEI-Basic support in MuseScore



Joint project between the **Digital and Cognitive Musicology Lab (DCML)** at the EPFL and the **RISM Digital Center** in Bern

- Research funding from the DCML
- In-kind contribution from the RISM Digital Center

Implementation of both import and export support within MuseScore with no plugin



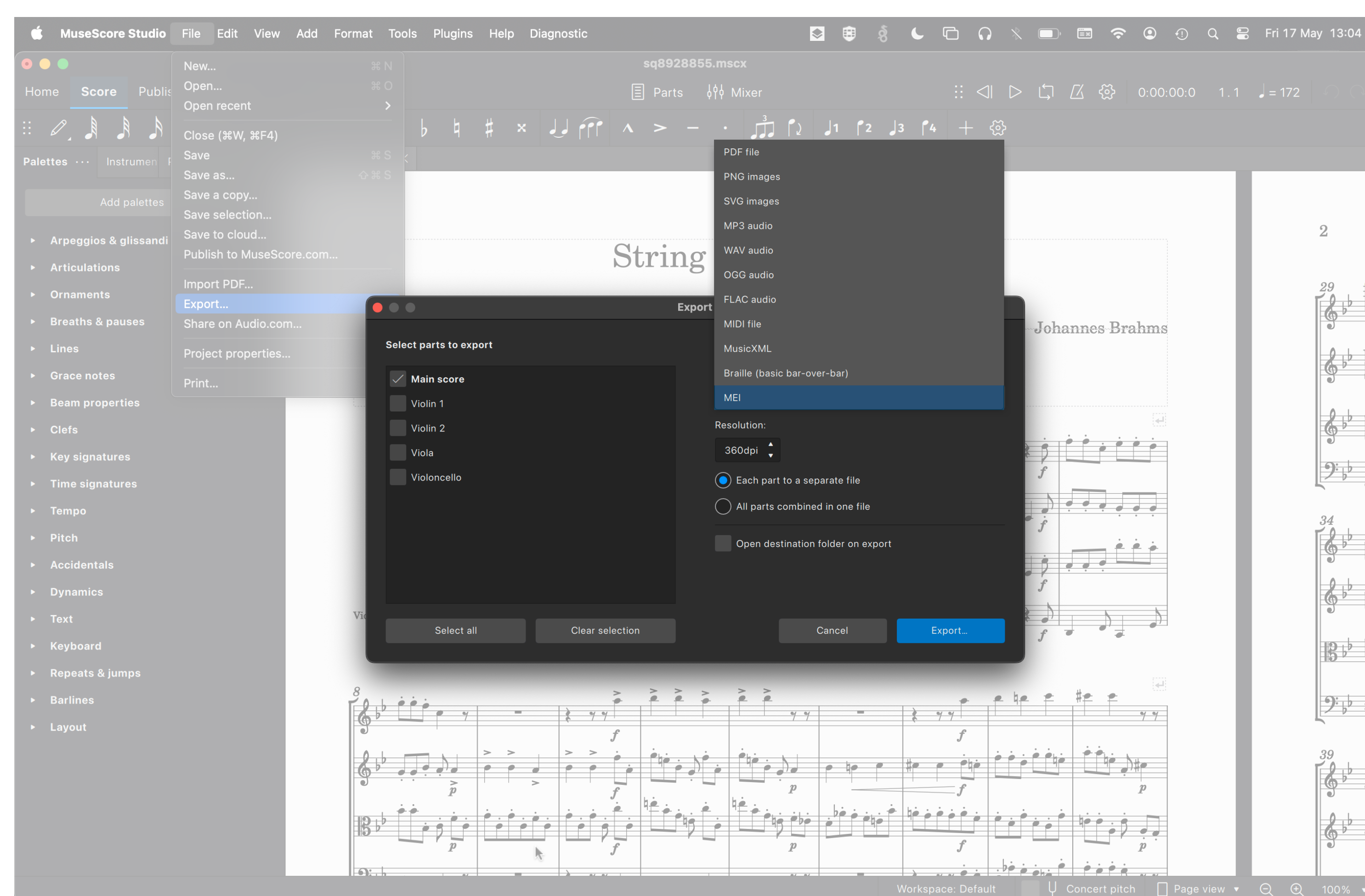
MEI-Basic tailored for data interchange

- Limits the ways a feature can be expressed and encoded
- Removal of modules (non-CWMN, editorial markup, etc.)
- Very basic metadata header

Features available

- General score structure
- Instrument labels
- Transposing information
- Measure numbering
- Time signatures, key signatures, and clefs
- Notes, grace notes, and rests
- Tuplets and beaming
- Ending and repeat signs
- Tempo markings
- Dynamics (ff, s.p, sfz etc.)
- Fermata
- Figured bass and chord labels
- Additional text directives
- Slurs and ties
- Note articulation (stacc.)
- Ornaments such as mordents and turns
- Ottava lines
- Hairpins

Import and export integrated into MuseScore



LibMEI for C++ code generation from the MEI-Basic Schema

MEI data type

data.ACCIDENTAL.WRITTEN.basic

Written standard accidental values.

Module

MEI

Used by

Data Types: These other Data Types reference data.ACCIDENTAL.WRITTEN.basic

data.ACCIDENTAL.WRITTEN: Written standard accidental values.

Allowed Values

s: Sharp.
f: Flat.
x: Double sharp (written as 2 sharps).
w: Double flat (written using crook).
tr: Triple sharp (written as a crook followed by a sharp).
ts: Triple sharp (written as a sharp followed by a crook).
tf: Triple flat.
n: Natural.
ns: Natural + flat used to cancel preceding double flat.
ss: Natural + sharp used to cancel preceding double sharp.

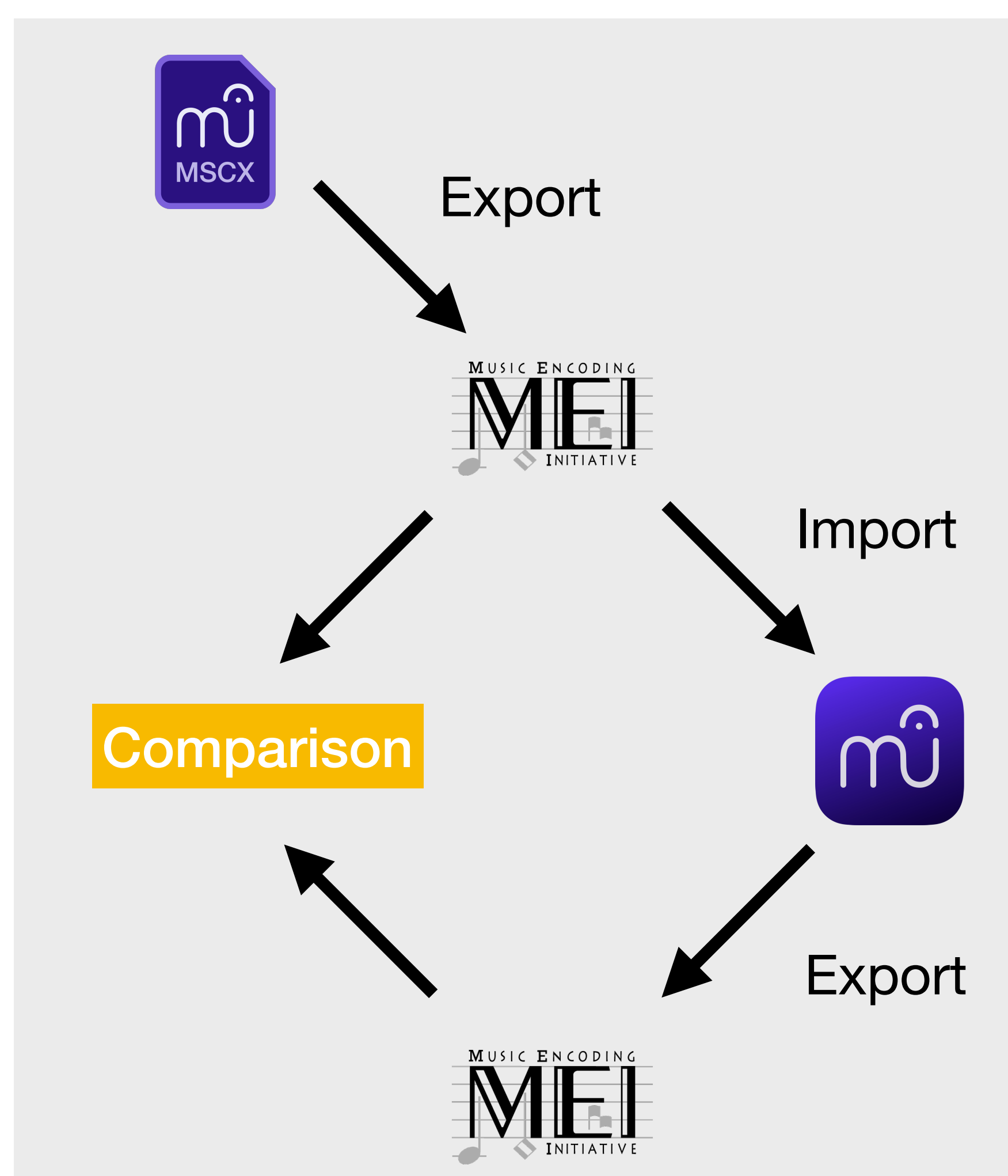
Declaration

```
<accid:accid="data.ACCIDENTAL.WRITTEN.basic" type="MEI">  
<desc>accid="MEI" written standard accidental values.</desc>  
<content>  
  <all>  
    <accid="s">Sharp</accid>  
    <accid="f">Flat</accid>  
    <accid="x">Double sharp (written as 2 sharps)</accid>  
    <accid="w">Double flat (written using crook)</accid>  
    <accid="tr">Triple sharp (written as a crook followed by a sharp)</accid>  
    <accid="ts">Triple sharp (written as a sharp followed by a crook)</accid>  
    <accid="tf">Triple flat</accid>  
    <accid="n">Natural</accid>  
    <accid="ns">Natural + flat used to cancel preceding double flat</accid>  
    <accid="ss">Natural + sharp used to cancel preceding double sharp</accid>  
  </all>  
</content>  
</accid:accid>
```

MuseScore methods using C++ data types

```
engraving::AccidentalType Convert::accidFromMEI(const libmei::data_ACCIDENTAL_WRITTEN meiAccid, bool& warning)  
{  
  warning = false;  
  switch (meiAccid) {  
    case (libmei::ACCIDENTAL_WRITTEN_NONE): return engraving::AccidentalType::NONE;  
    case (libmei::ACCIDENTAL_WRITTEN_n): return engraving::AccidentalType::NATURAL;  
    case (libmei::ACCIDENTAL_WRITTEN_f): return engraving::AccidentalType::FLAT;  
    case (libmei::ACCIDENTAL_WRITTEN_ff): return engraving::AccidentalType::FLAT2;  
    case (libmei::ACCIDENTAL_WRITTEN_tf): return engraving::AccidentalType::FLAT3;  
    case (libmei::ACCIDENTAL_WRITTEN_s): return engraving::AccidentalType::SHARP;  
    case (libmei::ACCIDENTAL_WRITTEN_x): return engraving::AccidentalType::SHARP2;  
    case (libmei::ACCIDENTAL_WRITTEN_ts): return engraving::AccidentalType::SHARP3;  
    case (libmei::ACCIDENTAL_WRITTEN_nf): return engraving::AccidentalType::NATURAL_FLAT;  
    case (libmei::ACCIDENTAL_WRITTEN_ns): return engraving::AccidentalType::NATURAL_SHARP;  
    case (libmei::ACCIDENTAL_WRITTEN_ss): return engraving::AccidentalType::SHARP_SHARP;  
    default:  
      warning = true;  
      return engraving::AccidentalType::NATURAL;  
  }  
}  
  
libmei::data_ACCIDENTAL_WRITTEN Convert::accidToMEI(const engraving::AccidentalType accid)  
{  
  switch (accid) {  
    case (engraving::AccidentalType::NONE): return libmei::ACCIDENTAL_WRITTEN_NONE;  
    case (engraving::AccidentalType::NATURAL): return libmei::ACCIDENTAL_WRITTEN_n;  
    case (engraving::AccidentalType::FLAT): return libmei::ACCIDENTAL_WRITTEN_f;  
    case (engraving::AccidentalType::FLAT2): return libmei::ACCIDENTAL_WRITTEN_ff;  
    case (engraving::AccidentalType::FLAT3): return libmei::ACCIDENTAL_WRITTEN_tf;  
    case (engraving::AccidentalType::SHARP): return libmei::ACCIDENTAL_WRITTEN_s;  
    case (engraving::AccidentalType::SHARP2): return libmei::ACCIDENTAL_WRITTEN_x;  
    case (engraving::AccidentalType::SHARP3): return libmei::ACCIDENTAL_WRITTEN_ts;  
    case (engraving::AccidentalType::NATURAL_FLAT): return libmei::ACCIDENTAL_WRITTEN_nf;  
    case (engraving::AccidentalType::NATURAL_SHARP): return libmei::ACCIDENTAL_WRITTEN_ns;  
    case (engraving::AccidentalType::SHARP_SHARP): return libmei::ACCIDENTAL_WRITTEN_ss;  
    default:  
      return libmei::ACCIDENTAL_WRITTEN_n;  
  }  
}
```

Round-trip testing



Good to know

- Available from MuseScore 4.2 through the user interface and through the command line
- Export and import of the layout (page and system begins) optional
- Use of the MEI @type attribute for some MuseScore features with no direct equivalent in MEI
- Various ways to handle the generation and preservation of MEI @xml:id (but with some limitations)

Some recommendations

Exporting from MuseScore

- Start from empty or “trusted” MuseScore files
- Be careful with heavily edited files (hidden elements, etc.)

Importing into MuseScore

- Use MEI-Basic - which can be exported from Verovio
- Keep in mind the use of @type attributes in MEI exported files
- Do your own round-trip evaluation



<https://music-encoding.org/musescore-doc/>

Feedback!

- MEI slack #musescore channel
- GitHub issues or PR - appreciated on the rism-digital/musescore fork